

Toward Patient-Specific Organoid Cultures: Evidence from Preclinical Cohorts

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Abstract

This paper reports a systematic evaluation of longitudinal electrode arrays. Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

1. Introduction

Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings argue for treating the two regimes separately in future analyses.

The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Residual error grows approximately linearly with the number of coupled parameters. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined.

2. Related Work

Sensitivity to the smoothing window proved modest across the range we examined. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings argue for treating the two regimes separately in future analyses. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

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3. Materials and Methods

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4. Experimental Setup

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5. Results

The measurements reported here were collected over twelve independent trials under controlled conditions. Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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6. Discussion

The reduced model retains most of the explanatory power at a fraction of the computational cost. Calibration drift over the trial period remained below the threshold set out in the protocol. Residual error grows approximately linearly with the number of coupled parameters. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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7. Limitations

We release the processing scripts and intermediate artefacts to support independent verification. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. The reduced model retains most of the explanatory power at a fraction of the computational cost.

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8. Conclusion

Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Sensitivity to the smoothing window proved modest across the range we examined.

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